

BUSINESS CYCLES. (BCs)

95

Macroeconomics

short-run fluctuations
in output

↓
business cycles

long-run changes in output

↓
economic growth

- Sometimes, economic conditions are very healthy

employment - high

output growth - good

business profits - high

price level - controlled

As happened in the US
in the 1990s

But, often these conditions are followed by the periods of crises

business failure

high unemployment

rising price level

happened in the US
during 2007-10

Further, the crises are sometimes mild (2001) & sometimes, severe (the great Depression)

These short-run fluctuations in output are known as business cycles.

- Keynes in the 1930s argued that forces of Aggregate Demand are primarily responsible for business cycles & also contributed on how governments can control such situations

◦ Formal definition

BCs are economy wide fluctuations

in total national o/p, income & employment,

usually lasting for a period of 2-10 years,

marked by expansion or contraction in

most sectors of the economy

← (A)

← (B)

← (C)

← (D)

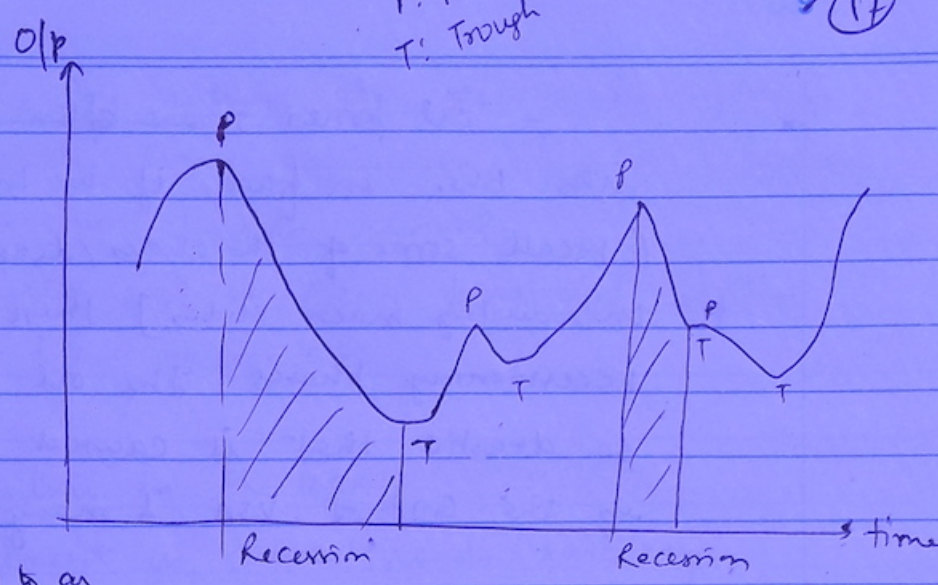
← (E)

- (A): economy wide fluctuations $\rightarrow$ occur throughout the economy & are not confined to a particular region or sector.
 For instance, it can't be the case that a $\downarrow$ in Delhi GDP (precisely SDP) without a consequent $\downarrow$ in GDP of India or State Domestic Product, other states is termed as BC.
 a counterpart of GDP, which is a term usually used for nations (also see pt. E for sectors)
- (B): more than one macro-variables are affected during the BCs, the effect is not just confined to say output
- (C): the $\downarrow$ or $\uparrow$ should persist for some period. While there are no definite benchmarks or thresholds to define a BC, but usually an $\uparrow$ or $\downarrow$ sh'd go beyond a few quarters
- (D): Typically, there are two phases - expansion
 or
 Recession / downturn
 $\downarrow$
 [As per the US NBER, if the o/p declines for 2 (more successive quarters)]
 also, see the diagram below / p. 17.
- (E): Most sectors of economy are affected.
 The impact is not on any one sector but usually all sectors (primary, secondary, tertiary) are affected.
 Hence BCs are pervasive in the entire economy
 also see (A) & (B) above.

P: Peak
T: Trough

- Peaks & Troughs are turning points

BCs need not be regular or periodic. The term cycle is actually a misnomer here.



A large recession is often referred to as a depression.
↓
(both in terms of amplitude & duration)

NO two BCs are similar

It is very difficult to capture & predict BCs.

- Why do BCs occur?

BC Theories $\begin{cases} \text{Exogenous} \\ \text{Internal} \end{cases}$

- Exogenous - Sources of BCs lie outside the economic system
changes in factors outside the economic system

Examples - Wars & Revolution

- elections - existence of political budget cycles

↓
do you expect (or actually see) that governments often spend more in election years or in the years preceding elections on various activities such as roads. This leads to political budget cycles.

- oil prices : ~~are often associated~~ are often associated with BCs. In fact, if we look at major oil shocks [recall some of the slides/diagrams we discussed during introductory macro slides] there have been associated with recessionary phases. The oil price shock of 1973 was so drastic that it caused a "structural change" in the GDP of USA & many other countries of the world.

- population migrations - recall from the micro theory that $Y = f(K, L)$, if there are large scale population migrations, then can affect Y (the output)

- Discovery of new resources - can affect the potential GDP of any country & in turn affect the actual GDP.

- scientific breakthroughs / technological innovations

- Natural disasters & epidemics.

Consider the Covid-19 pandemic, for instance
see some screenshots on the next two pages.

According to the International Monetary Fund, 'the global economy' is already into a recessionary phase:

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IMF chief: 'Clear we have entered recession'

AFP | Mar 27, 2020, 09:31 PM IST



WASHINGTON: The coronavirus pandemic has driven the global economy into a downturn that will require massive funding to help developing nations, IMF chief Kristalina Georgieva said on Friday.

"It is clear that we have entered a recession" that will be worse than in 2009 following the global financial crisis, she said in an online press briefing.

With the worldwide economic "sudden stop," Georgieva said the fund's estimate "for the overall financial needs of emerging markets is \$2.5 trillion." But she warned that "we believe this is on the lower end." Over 80

countries already have requested emergency aid from the International Monetary Fund.

A Financial Times report indicated the same for the UK:

The UK economy is plunging into a deeper recession than the 2008-09 financial crisis, according to the most reliable data published so far, with unemployment surging and the public finances sliding sharply into the red because of the coronavirus pandemic.

Economists said there was huge uncertainty over the depth of the coming recession and the speed of the subsequent recovery. Some predict the downturn will be short and sharp, but fear that like a decade ago, there will also be a painful transition to a persistently weaker path for economic growth.

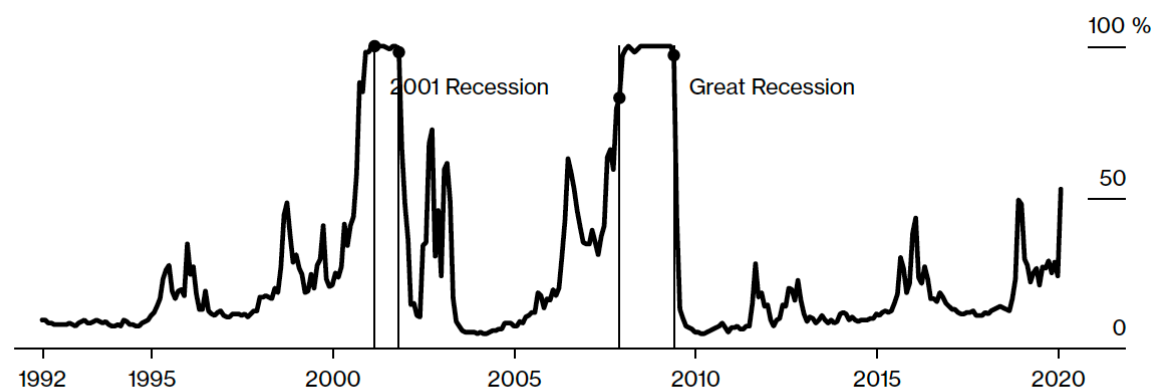
Early this month, a Bloomberg report (partial screenshot pasted below) indicated significant chances of recession for the US economy. This report was dated 11th March, when the Covid-19 cases in the US were around 1000. With a substantially rise in number of cases (approximately 180,000 as on 01 April) and economy activity virtually halted, the US has with almost certainty entered recession.

Bloomberg Economics created a model to determine America's recession odds. The chance of a recession within the next year now stands at 53%, the highest reading since the U.S. exited the Great Recession in June 2009 and significantly higher than the 24% seen in the prior month. The latest update combines February reports, estimates for other figures not yet released for the month, and financial markets data from early March.

[Sign up](#) to receive monthly updates to this recession tracker by email.

Risks Mount

Probability of U.S. recession within 12 months



Source: Bloomberg Economics

- Internal - Sources of BCs lie within the economic system
In this approach, every expansion breeds recession
& every contraction breeds revival

Within economic system, it is often financial sector where many BCs have originated

examples ◦ In the 19th Century in the USA,
heavy investment in canals/roads/railways;
over-construction of railways
people took too much debt
banking sector crisis & the impact on o/p & prices

- In the 1990s (New Economy Bubble)
huge investment in ICTs - people invested heavily in
software companies, the internet, etc

At one time about 1/5th of the US GDP came from
these investments

Eventually the investors became skeptical about the
fundamental value of many of these firms
boom & fall in stock prices; o/p & prices

- Financial crisis (2007-10) / Housing Bubble

subprime lending

Aggregate Demand (AD) & Business Cycles (BCs)

(20)

- Why study AD as part of BCs
as discussed earlier, most explanations of BCs lie in AD

AD:- is the total quantity of output bought at a given level of prices, ceteris paribus.

four components

(i) Consumption

$$C = f(Y_d) \quad \begin{array}{l} \nearrow \text{disposable} \\ \text{income} \end{array}$$

(the consp. function)

$$Y_d = Y - T \quad \begin{array}{l} \nwarrow \text{Income} \\ \rightarrow \text{taxes} \end{array}$$

But, we also discussed that future income & wealth (or Assets) also affect C.

$$\text{Then, } C = f(Y_d, Y_f, A)$$

+ + +

(ii) Investment

$$I = I(r)$$

In addition to r , it depends on $\begin{array}{l} \nearrow \text{taxes} \\ \searrow \text{expectations abt future} \end{array}$

(iii) Government Expenditure

$$G = G_0 \rightarrow \begin{array}{l} \nearrow \text{exogenous in nature} \\ \nwarrow \text{considered as given} \end{array}$$

(iv) Net Exports

$$NX \equiv \underbrace{X}_{\text{exports}} - \underbrace{M}_{\text{imports}}$$

$$X = f(Y^*, \text{relative prices, exchange rate})$$

$$M = f(Y, \text{relative prices, exchange rate})$$

$Y^* \rightarrow$ income of the foreign country

$Y \rightarrow$ domestic income

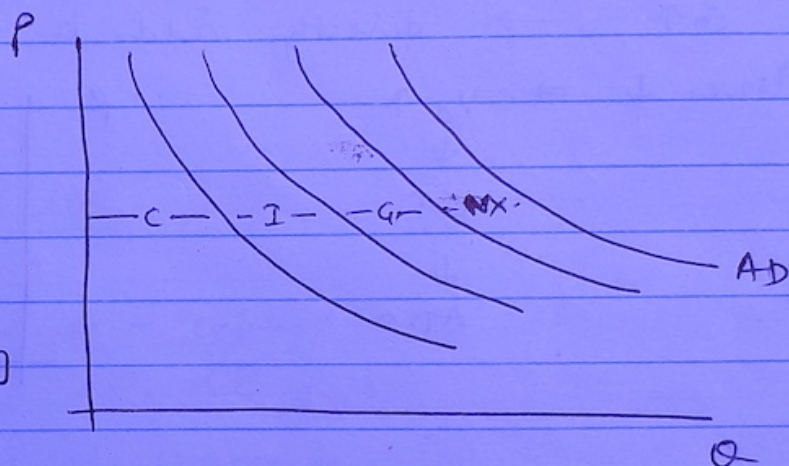
Note that your imports are determined by your capacity to buy
[i.e. your income]

& your exports are a function of the income of the country
that buys your products i.e. foreign country (i.e. Y^*)

$$AD = C + I + G + NX$$

Note that AD is
downward sloping

[see chap 22 in
Samuelson's book]



shifts in AD

Policy variables

↑
govt can influence

Exogenous

(i.e.
determined outside
the AD-AS
framework)

Monetary policy

fiscal policy

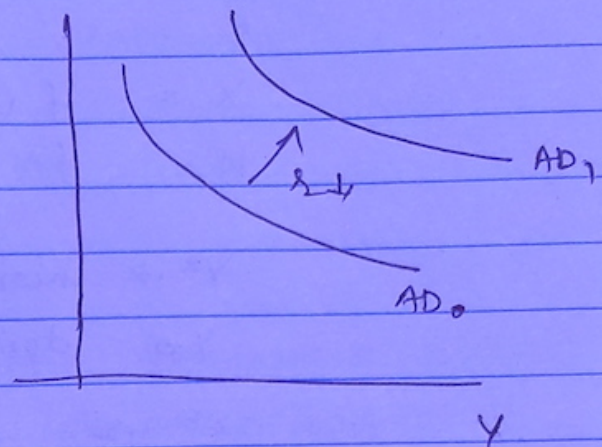
8 Policy (i) Monetary policy

for instance, decline in the interest rate (r) or monetary expansion

$\Rightarrow$ better credit conditions & higher investment

(Note: $I = I(r)$)

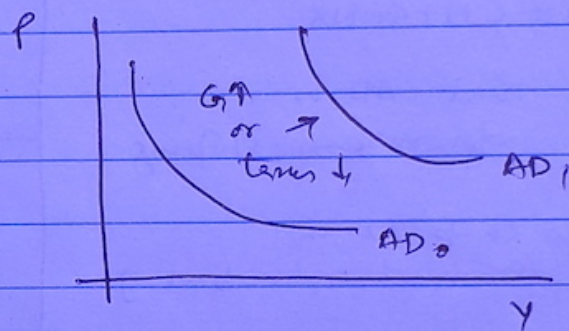
$\Rightarrow AD \uparrow$



(ii) Fiscal policy As you know, fiscal policy is about G (govt exp.) & T (taxes)

$G \uparrow \rightarrow$ directly adds to $AD \rightarrow AD \uparrow$

taxes $\downarrow \Rightarrow Y_d \uparrow$
 $\downarrow$
 $C \uparrow$
 $\downarrow$
 $AD \uparrow$



Exogenous

- $\uparrow$ in foreign output $\Rightarrow NX \uparrow \rightarrow AD \uparrow$

- $\uparrow$ in wealth (say, stock market $\uparrow$) $\Rightarrow C \uparrow$

- Technological advances - new opportunities for investment $\Rightarrow I \uparrow \rightarrow AD \uparrow$

- Other miscellaneous factors (political, climatic, oil prices, demonetisation, Covid-19) also affect AD

Business cycles in last 100-150 years

Earlier → prolonged crisis & cumulative slumps

Now → milder & less severe

This is primarily due to

- a larger & more stable government sector
- better understanding of macroeconomics
[see the introductory macro slides;
contribution of Keynes; better use
of fiscal & monetary policies by
govts to control their BCs]

MULTIPLIER

- If expenditure (say, consumption expenditure) changes by one unit, by what amount does the output change?
quantification of the change in expenditure on output
- helpful for policy - governments can use this to calculate the amount of expenditure that ^{is} needed in order to push an economy out of recession, for example.
(by ↑ of GDP)
- The multiplier is the impact of one unit change in exogenous expenditure on total output.

$$Y = C + I + G + NX$$

Consider a simple model

$$Y = C + I \quad (\text{ie. we assume } G=0 \text{ \& } NX=0)$$

adding G to the equation
does not affect our
conclusions though.

closed
economy

$$\therefore C = \alpha + \beta Y$$

$$\Rightarrow Y = \alpha + \beta Y + I$$

$$\Rightarrow (1 - \beta) Y = \alpha + I$$

$$\Rightarrow \boxed{\frac{\partial Y}{\partial I} = \frac{1}{1 - \beta}}$$

LHS is the change in o/p per unit change in investment expenditure.

$$\therefore C = \alpha + \beta Y \quad \& \quad \frac{\partial C}{\partial Y} = \beta$$

β is Marginal propensity to consume

$$\text{Thus, } \frac{\partial Y}{\partial I} = \frac{1}{1 - MPC} = \frac{1}{MPS}$$

$$\text{If } MPC = 2/3 ; \quad \frac{\partial Y}{\partial I} = \frac{1}{1 - 2/3} = 3$$

Thus, multiplier is 3 or an one unit change in investment expenditure leads to 3 times change in output.

Intuition underlying multiplier

- construction of woodsheds

(see the screenshots from Samuelson's book)

Similarly, you can see that the multiplier for govt expenditure G is also $\frac{1}{1 - \beta}$. These multipliers (govt. expenditure & investment) are known as expenditure multipliers.

Multiplier

Source: pp. 440-441, Samuelson's book

Woodsheds and Carpenters. Why is it that the multiplier is greater than 1? Let's suppose that I hire unemployed workers to build a \$1000 woodshed. My carpenters and lumber producers will get an extra \$1000 of income. But that is not the end of the story. If they all have a marginal propensity to consume of $\frac{2}{3}$, they will now spend \$666.67 on new consumption goods. The producers of these goods will now have extra incomes of \$666.67. If their *MPC* is also $\frac{2}{3}$, they in turn will spend \$444.44, or $\frac{2}{3}$ of \$666.67 (or $\frac{2}{3}$ of $\frac{2}{3}$ of \$1000). The process will go on, with each new round of spending being $\frac{2}{3}$ of the previous round.

Thus an endless chain of *secondary consumption spending* is set in motion by my *primary* investment of \$1000. But, although an endless chain, it is an ever-diminishing one. Eventually it adds up to a finite amount.

Using straightforward arithmetic, we can find the total increase in spending in the following manner:

$$\begin{array}{rcl}
 \begin{array}{r}
 \$1000.00 \\
 + \\
 666.67 \\
 + \\
 444.44 \\
 + \\
 296.30 \\
 + \\
 197.53 \\
 + \\
 \vdots \\
 \hline
 \$3000.00
 \end{array}
 & = &
 \begin{array}{r}
 1 \times \$1000 \\
 + \\
 \frac{2}{3} \times \$1000 \\
 + \\
 (\frac{2}{3})^2 \times \$1000 \\
 + \\
 (\frac{2}{3})^3 \times \$1000 \\
 + \\
 (\frac{2}{3})^4 \times \$1000 \\
 + \\
 \vdots \\
 \hline
 \frac{1}{1 - \frac{2}{3}} \times \$1000, \text{ or } 3 \times \$1000
 \end{array}
 \end{array}$$

This shows that, with a *MPC* of $\frac{2}{3}$, the multiplier is 3; it consists of the 1 of primary investment plus 2 extra of secondary consumption responding.

The same arithmetic would give a multiplier of 4 for a *MPC* of $\frac{3}{4}$, because $1 + \frac{3}{4} + (\frac{3}{4})^2 + (\frac{3}{4})^3 + \dots$ eventually adds up to 4. For a *MPC* of $\frac{1}{2}$, the multiplier would be 2.¹

The size of the multiplier thus depends upon how large the *MPC* is. It can also be expressed in terms of the twin concept, the *MPS*. For a *MPS* of $\frac{1}{4}$, the *MPC* is $\frac{3}{4}$ and the multiplier is 4. For a *MPS* of $\frac{1}{3}$, the multiplier is 3. If the *MPS* were $1/x$, the multiplier would be x .

By this time it should be clear that the simple multiplier is always the inverse, or reciprocal, of the marginal propensity to save. It is thus equal to $1/(1 - MPC)$. Our simple multiplier formula is

$$\begin{aligned}
 \text{Change in output} &= \frac{1}{MPS} \times \text{change in investment} \\
 &= \frac{1}{1 - MPC} \times \text{change in investment}
 \end{aligned}$$